CUSTOMER SPENDING PREDICTION

Machine Learning Project Report

Prepared using the supplied customer dataset

Dataset size: 1,000 records | Features used: 6 | Target: spending

1. Abstract

This project develops a supervised regression workflow to estimate customer spending using demographic and purchase-behaviour information. The workflow includes data cleaning, exploratory data analysis, preprocessing of numeric and categorical variables, comparison of regression models, evaluation on a held-out test set, and a Streamlit dashboard for interactive predictions.

2. Objectives

• Inspect and clean the supplied dataset.

• Explore the distribution of spending and its relationship with selected attributes.

• Train and compare multiple regression algorithms.

• Evaluate models with MAE, RMSE and R^2.

• Provide an interactive dashboard for demonstration.

3. Dataset Description

The dataset contains 1000 rows and 8 columns. The target variable is **spending**. Predictors used are: age, gender, education, income, country, purchase_frequency. The **name** field is excluded from modeling because names are identifiers and should not be used as predictive inputs.

Item	Value
Rows in original file	1000
Rows after duplicate removal	1000
Columns	8
Missing cells after cleaning	0
Duplicate rows removed	0
Mean spending	9,613.30
Median spending	9,430.40
Minimum spending	611.99
Maximum spending	25,546.50

4. Data Cleaning and Preparation

Text fields are trimmed, blank text values are treated as missing, and duplicate rows are removed. Missing numeric values are imputed with the median; missing categorical values are imputed with the most frequent category. Categorical variables are one-hot encoded. The data is split into 80% training and 20% test sets using random_state=42 to make the experiment reproducible.

5. Exploratory Data Analysis

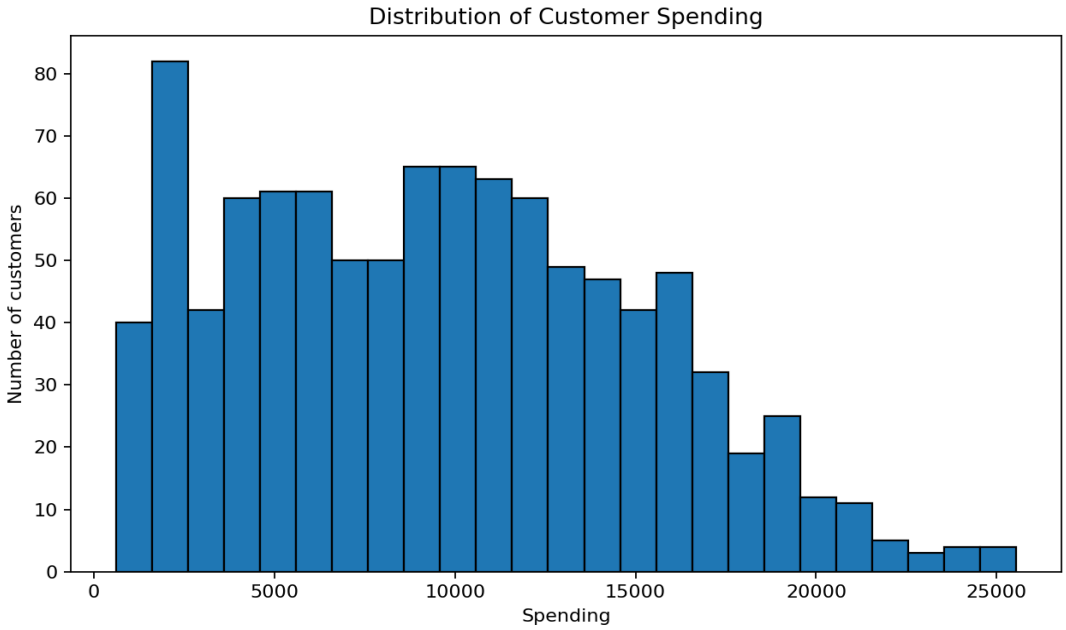


Figure 1. Distribution of customer spending.



Figure 2. Income versus spending.

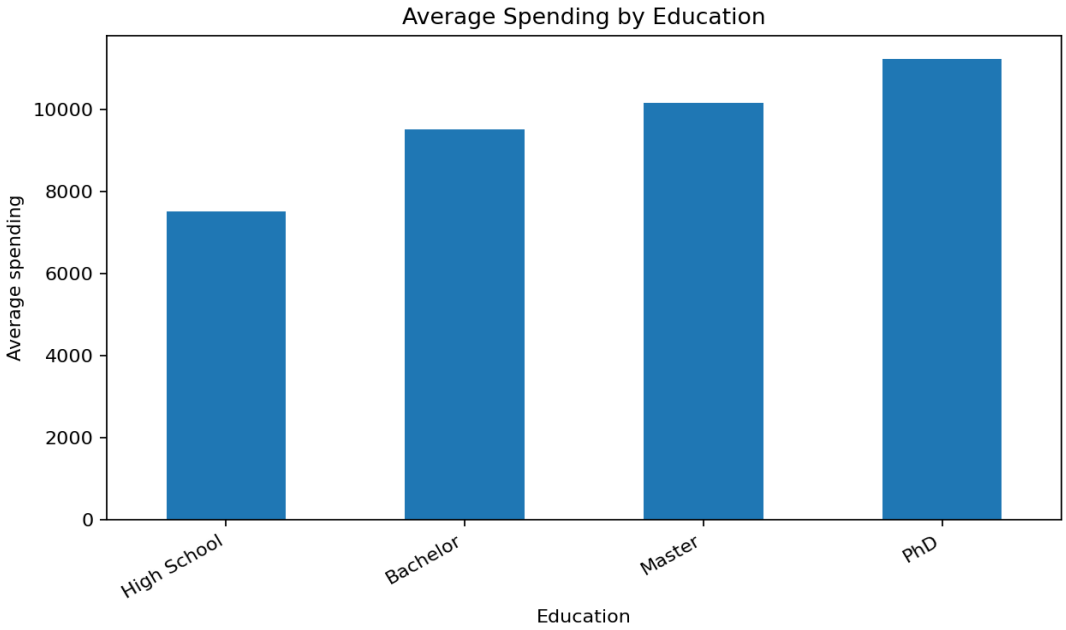


Figure 3. Average spending by education category.

6. Modeling Methodology

The target is a continuous numeric value, so regression algorithms are used. The project compares Linear Regression, Ridge Regression, Random Forest, Extra Trees, and HistGradientBoosting. The preprocessing and estimator are packaged in a single scikit-learn Pipeline so that the same transformations are applied consistently at training and prediction time.

7. Results

The lowest-RMSE model on this run is **Hist Gradient Boosting**. Test-set metrics are shown below. MAE and RMSE are measured in the same units as spending; lower values are better. R^2 describes the fraction of target variation explained relative to a constant-mean baseline and can be negative when predictions perform poorly.

Model	MAE	RMSE	R^2
Hist Gradient Boosting	269.01	385.76	0.995
Random Forest	589.30	777.33	0.980
Extra Trees	594.95	830.72	0.977
Ridge Regression	1,007.12	1,318.98	0.942
Linear Regression	1,097.94	1,439.45	0.931

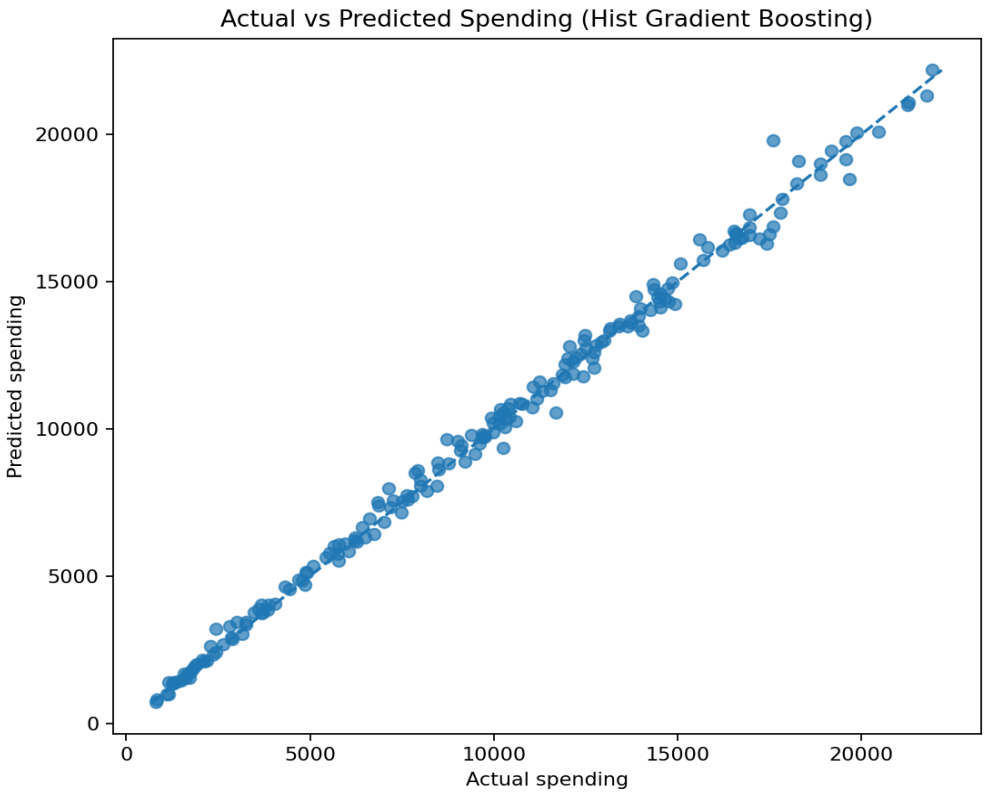


Figure 4. Actual versus predicted spending for the selected model.

8. Dashboard and Usage

The Streamlit dashboard includes three tabs: interactive prediction, dataset exploration, and model performance. Run `streamlit run app.py` after installing the packages and ensuring the saved model exists. Run `python train_model.py` to retrain the models.

9. Limitations and Future Work

• The dataset may not represent real customers or future populations; validate its provenance before drawing business conclusions.

• The meaning and unit of purchase_frequency should be confirmed from the dataset documentation.

• The model learns associations, not causal effects.

• Future improvements include cross-validation, tuning, robust outlier analysis, explainability, and testing on a truly unseen dataset collected at a later time.

10. Conclusion

A complete educational workflow has been assembled from data preparation through model evaluation and an interactive demo. The best model is selected by test-set RMSE, while the comparison table supports transparent model selection. The project should be treated as a prototype until data quality and real-world validity are independently verified.